

FitBuddy

AI Fitness Plan Generator using
Gemini Models

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Abstract

FitBuddy is an AI-based web application that creates personalised fitness plans in seconds. The user enters basic details such as age, height, weight, goal, fitness level, available equipment and food preference. FitBuddy converts these details into a structured prompt and sends it to a Google Gemini model.

The model returns a weekly workout schedule, a daily meal plan and simple health tips. The application shows the plan in a clean layout and lets the user download it.

The project removes the need for costly trainers and generic online plans. It shows how generative AI can be used for practical, everyday health support.

Keywords

Generative AI · Google Gemini · Prompt Engineering · Fitness Planning · Personalisation · Python · Streamlit

1. Introduction

Regular exercise and balanced food are the base of good health. Most people, however, do not know what to do, how much to do, or how to adjust a plan to their own body and lifestyle.

Large Language Models (LLMs) can understand a person's details and write a clear, customised plan. FitBuddy uses Gemini models to do this in a simple chat-free form: fill a form, get a plan.

Why this project?

- **Need:** Personal trainers and dieticians are expensive and not available to everyone.
- **Gap:** Generic plans on the internet ignore age, injuries, equipment and food habits.
- **Opportunity:** Gemini can produce structured, personalised content quickly at low cost.

In one line

FitBuddy = user profile + smart prompt + Gemini model = a personal fitness plan.

2. Problem Statement

People who start fitness journeys often quit because their plan is not suited to them. The main problems are:

Problem	Impact
One-size-fits-all plans	Too hard for beginners, too easy for advanced users
High cost of trainers	Not affordable for students and many families
No local food awareness	Diet plans do not match Indian or regional food habits
Time and equipment limits	Plans need a gym; many users train at home
Lack of guidance	No one to explain exercises or adjust the plan

Problem statement

To build an AI-powered system that generates safe, personalised, easy-to-follow workout and diet plans from a few user inputs using Gemini models.

3. Objectives and Scope

Objectives

- Collect user health and fitness details through a simple form.
- Generate a personalised weekly workout plan.
- Generate a matching daily diet plan based on food preference.
- Present the output in a clear, structured format.
- Keep the system low-cost, fast and easy to use.

Scope

In scope	Out of scope
Workout and diet plan generation	Medical diagnosis or treatment
Beginner to advanced levels	Live tracking with wearables (future work)
Home and gym workouts	Real-time video form correction
Vegetarian, non-vegetarian and vegan diets	Prescribing supplements or medicines

4. Existing vs Proposed System

Existing system

- Fixed plans in PDFs, blogs and videos.
- Fitness apps with paid subscriptions.
- Manual advice from trainers, which is costly and time-bound.

Proposed system: FitBuddy

- Plan generated from the user's own details.
- Uses Gemini for natural, structured and adaptable output.
- Free to run on the Gemini API free tier for small usage.

Comparison

Feature	Existing	FitBuddy
Personalisation	Low	High
Cost	Medium to high	Low
Time to get plan	Hours to days	Seconds
Food preference support	Limited	Yes
Easy to update plan	No	Yes, regenerate anytime

5. Gemini Models Overview

Gemini is Google's family of multimodal AI models. It can understand text, images, audio and more, and it can produce well-structured text such as tables and JSON. FitBuddy uses Gemini through the Gemini API.

Why Gemini for FitBuddy

Capability	Use in FitBuddy
Strong instruction following	Follows the plan format we ask for
Structured (JSON) output	Easy to display plans as tables and cards
Fast Flash-type models	Quick response and low cost
Long context window	Can hold profile, rules and previous plan together
Multimodal input	Future: analyse a meal photo or workout image

Model choice

A fast, low-cost Gemini Flash model is used for normal plan generation. A more advanced Pro model can be used when deeper reasoning is needed. Always use the latest model name shown in Google AI Studio.

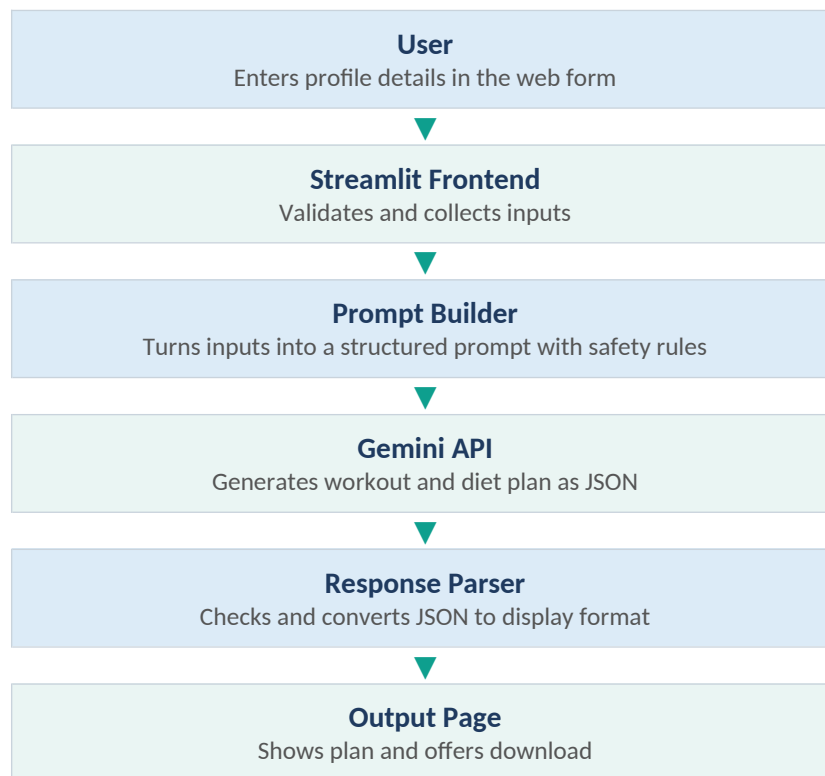
6. Technology Stack

Layer	Technology	Purpose
AI Model	Google Gemini (via Gemini API)	Generates plans
Language	Python 3	Core application logic
Frontend	Streamlit	Simple web interface
SDK	google-generativeai	Calls the Gemini API
Data handling	JSON, Pandas	Parse and show plans
Export	PDF / Markdown download	Save the plan
Hosting	Streamlit Community Cloud	Free deployment
Tools	VS Code, Google AI Studio, GitHub	Development and version control

API keys are stored as environment variables or in Streamlit secrets, never in the source code.

7. System Architecture

FitBuddy follows a simple pipeline. Each layer has one clear job.



8. Modules Description

#	Module	Description
1	User Input	Form for age, gender, height, weight, goal, level, equipment, diet, days per week
2	BMI and Profile	Calculates BMI and a simple profile summary
3	Prompt Builder	Creates the prompt with role, rules and output format
4	Gemini Connector	Sends the prompt and receives the response
5	Response Parser	Reads JSON, handles errors and missing fields
6	Plan Display	Shows workout table, meal table and tips
7	Export	Downloads the plan as PDF or text

9. Workflow

Step	Action	Result
1	User opens FitBuddy and fills the form	Profile data collected
2	App checks inputs (valid age, height, weight)	Bad data rejected
3	BMI is calculated	Profile summary ready
4	Prompt is built with the profile and safety rules	Final prompt ready
5	Prompt is sent to the Gemini model	Plan returned as JSON
6	JSON is parsed and validated	Clean plan data
7	Plan is displayed and can be downloaded	User gets the plan

Error handling

If the API fails or returns invalid JSON, FitBuddy retries once and then shows a friendly message.

10. Prompt Design

Good prompts give good plans. FitBuddy's prompt has four parts:

Part	What it does
Role	Tells the model to act as a certified fitness and nutrition coach
Context	Passes the user's profile values
Rules	Safe exercises, no medical claims, respect food preference and equipment
Output format	Asks for strict JSON with fixed keys

Prompt template

```
You are a certified fitness and nutrition coach.  
Create a {days}-day weekly plan for this user:  
Age: {age}, Gender: {gender}, BMI: {bmi}  
Goal: {goal}, Level: {level}  
Equipment: {equipment}, Diet: {diet}
```

Rules:

- Keep exercises safe for the given level.
- Do not give medical advice.
- Match the diet preference.
- Reply ONLY in the JSON format below.

11. Input and Output Format

User inputs

Field	Type	Example
Age	Number	21
Gender	Choice	Male / Female / Other
Height and Weight	Number	170 cm, 68 kg
Goal	Choice	Weight loss, muscle gain, stay fit
Fitness level	Choice	Beginner / Intermediate / Advanced
Equipment	Choice	No equipment / Dumbbells / Full gym
Diet	Choice	Veg / Non-veg / Vegan
Days per week	Number	4

Model output (JSON)

```
{
  "summary": "Short plan overview",
  "workout": [{"day": "Day 1", "focus": "Upper body",
    "exercises": [{"name": "Push-ups", "sets": 3, "reps": "10-12"}]},
  "diet": [{"meal": "Breakfast", "items": "Idli, sambar, fruit"}],
  "tips": ["Drink 2-3 litres of water"]
}
```

12. Implementation

The core of FitBuddy is a short Python function that sends the prompt to Gemini and reads the JSON reply.

```
from google import genai
import json, os

client = genai.Client(api_key=os.environ["GEMINI_API_KEY"])

def generate_plan(prompt: str) -> dict:
    response = client.models.generate_content(
        model="gemini-2.5-flash",          # use latest Flash model
        contents=prompt,
        config={"response_mime_type": "application/json"},
    )
    return json.loads(response.text)
```

- **response_mime_type** asks Gemini to return valid JSON only.
- **Environment variable** keeps the API key out of the code.
- **Streamlit** calls `generate_plan()` when the user clicks Generate.

13. User Interface Design

The interface is kept simple so anyone can use it in under a minute.

Screen	Elements
Home	App name, short tagline, Start button
Profile form	Sliders and dropdowns for all inputs, Generate Plan button
Plan page	Summary card, weekly workout table, meal table, tips
Download	Buttons to save the plan as PDF or text

Design principles

- Clean layout with a calm teal and navy colour theme.
- Big buttons and clear labels for mobile users.
- Loading indicator while the plan is being generated.
- Friendly messages for errors and empty inputs.

14. Sample Output

Sample profile: 21 years, male, beginner, goal muscle gain, no equipment, vegetarian, 4 days per week.

Weekly workout

Day	Focus	Exercises
Day 1	Upper body	Push-ups 3×10, Pike push-ups 3×8, Plank 3×30s
Day 2	Lower body	Squats 3×15, Lunges 3×10, Glute bridge 3×15
Day 3	Rest	Light walk and stretching
Day 4	Core and cardio	Mountain climbers 3×20, Leg raises 3×12, Skipping 5 min
Day 5	Full body	Burpees 3×8, Squats 3×12, Push-ups 3×10

Daily meals

Meal	Suggestion
Breakfast	Vegetable upma, curd, banana
Lunch	Rice, dal, vegetables, salad
Snack	Roasted chana, fruit
Dinner	Chapati, paneer curry, cucumber

Note: This is an example of the type of output. Actual plans vary with each user and each request.

15. Testing

Testing checked that inputs, prompts, model responses and the interface work correctly.

ID	Test case	Expected result	Status
T1	Valid profile submitted	Plan generated and displayed	Pass
T2	Empty or invalid age	Error message shown	Pass
T3	Vegetarian selected	No non-veg items in diet	Pass
T4	No equipment selected	Only bodyweight exercises	Pass
T5	Invalid JSON from API	Retry, then friendly error	Pass
T6	Missing API key	Clear configuration message	Pass
T7	Plan download	File saves correctly	Pass

Types of testing: functional testing, input validation testing, prompt testing and usability testing.

16. Results and Discussion

- **Speed:** A complete plan is generated within a few seconds.
- **Personalisation:** Plans change clearly with goal, level, equipment and diet.
- **Structure:** JSON output made the display reliable and easy to format.
- **Usability:** The form-based flow is simple, with no chat skills needed.

Key observations

Observation	Learning
Clear prompts give better plans	Role, rules and format must be specific
Strict JSON format reduces errors	Always validate the parsed response
Safety rules improve trust	Avoid extreme or risky advice

Outcome

FitBuddy meets its objective: a fast, simple and personalised fitness plan generator built on Gemini.

17. Advantages and Limitations

Advantages	Limitations
Personalised plans in seconds	Not a replacement for doctors or trainers
Low cost and easy access	Needs an internet connection and API access
Supports home and gym workouts	AI output can occasionally be inaccurate
Supports veg, non-veg and vegan diets	No live tracking or form correction yet
Easy to extend with new features	Free API tier has usage limits

18. Safety, Privacy and Ethics

Area	Measure taken
Health safety	Disclaimer on every plan: consult a doctor before starting
Prompt safety	Rules block extreme diets and unsafe exercise advice
Data privacy	No personal data stored; inputs used only for the request
API security	Keys kept in environment variables or secrets
Fairness	Plans avoid bias based on gender, body type or background
Transparency	Users are told the plan is AI-generated

Disclaimer

FitBuddy gives general fitness guidance only. People with health conditions, injuries or pregnancy should consult a qualified professional first.

19. Future Enhancements

Enhancement	Benefit
Progress tracking and charts	Motivates users and adapts plans
Wearable and Google Fit integration	Uses real activity and heart-rate data
Meal photo analysis with Gemini vision	Estimates calories from a photo
Regional language support (e.g., Tamil)	Reaches more users
Voice assistant	Hands-free workout guidance
Mobile app version	Easier daily use
Weekly auto-adjusted plans	Plan improves with user feedback

20. Conclusion

FitBuddy shows how generative AI can make healthy living easier. With a simple form, a well-designed prompt and a Gemini model, users receive personalised workout and diet plans within seconds.

The project uses easy tools, keeps cost low and includes safety measures. It can grow into a complete digital fitness companion with tracking, voice and multilingual support.

Final thought

Good health should be simple, personal and accessible. FitBuddy is a small step in that direction.

21. References

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